CAMPUS CONNECT

A University-verified marketplace for Buying, selling, Renting and Exchanging, student resources.



SOFTWARE REQUIREMENTS SPECIFICATION

Particular	Details
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1. Introduction

1.1 Purpose

Campus Connect is a university-oriented web platform intended to bring two student-facing experiences together in one interface: (1) a campus marketplace for discovering, listing, buying, selling, renting and exchanging reusable goods; and (2) Career Connect for discovering internships, full-time roles, capstone projects and mentorship opportunities, applying to opportunities and tracking application progress. This SRS specifies the behaviour and quality expectations of the target system.

1.2 Intended audience

- Project team members responsible for requirements, UI, application logic, database and testing.
- Faculty guide and academic evaluators reviewing the project scope, completeness and engineering process.
- Students and campus members using the marketplace and career discovery functions.
- Recruiters or company representatives publishing opportunities and reviewing applications.
- Administrators responsible for user verification, moderation, platform configuration and operational oversight.

1.3 Document conventions

Requirement identifiers use FR-SH for shared functions, FR-MKT for marketplace functions, FR-CAR for career functions, FR-ADM for administration, NFR for quality requirements, DATA for data constraints and INT for interfaces. Each requirement is intended to be individually testable. Verification methods are inspection, demonstration, analysis or test.

1.4 Product vision

The product vision is to make campus resources easier to share and career opportunities easier to reach. Campus Connect reduces fragmentation by providing a consistent navigation, account experience, dashboard, notifications and theme preference across both modules. The marketplace supports circular reuse and affordability; Career Connect supports structured opportunity discovery and application tracking.

1. Introduction – Scope and References

1.5 Product scope

The initial academic product is a responsive web prototype. The current prototype demonstrates client-side navigation, marketplace listing discovery and management, favourites, career opportunity discovery, recruiter role management, student applications, application status tracking, dashboards, notifications and light/dark theme preferences. The current build stores records in browser localStorage and uses seeded demonstration content. It is not yet a shared, production-ready multi-user service.

1.6 In-scope target capabilities

- Shared account registration, login/logout, profile and role-aware dashboard.
- Marketplace listings, search, category/condition/price filtering, sorting, favourites, seller enquiries and listing ownership controls.
- Career opportunities, recruiter role postings, student applications, recruiter applicant review and status updates.
- In-app notification centre for relevant marketplace and career events.
- Responsive interface, accessible forms, feedback states and light/dark theme.
- Target-state server-side persistence, university/company verification, secure access control, image upload, moderation, messaging and deployment readiness as planned enhancements.

1.7 Out of scope unless approved

- A guaranteed payment gateway, escrow, refunds or payment dispute resolution.
- Shipping, courier, delivery fleet or physical pickup scheduling.
- Native iOS/Android applications.
- Automated university SIS/ERP integration or guaranteed single sign-on.
- AI-based price recommendations, automated candidate ranking or recruitment decisions.
- Cross-university federation in the first campus pilot.

1.8 References

Reference	How used
Campus Connect Lean Canvas 2.0 (10-Nov-2026)	Primary reference for marketplace problem, solution, users, value proposition, metrics, channels, costs, revenue assumptions and staged expansion.
Campus Connect Tech Conference Final presentation	Primary reference for the two-module product, prototype features, browser-local persistence and known limitations.
Campus Connect HTML prototype description / project artifact	Reference for the actual single-file HTML/CSS/JavaScript implementation and user-facing workflows.
Earlier Campus Marketplace SRS	Structural reference for SRS organization and requirement traceability; its marketplace-only scope is not substituted for the current two-module product.
IEEE Std 830-1998	Historical SRS structure reference; this document is a project-specific adaptation.
ISO/IEC 25010	Informative quality model reference for usability, reliability, security, performance and maintainability.
WCAG 2.1	Accessibility design and evaluation reference.

2. Overall Product Description

2.1 Product perspective

Campus Connect is a standalone campus web application. The current demonstration is a single HTML file with embedded CSS and JavaScript. It runs in a browser and stores state in localStorage. The target system is expected to evolve into a client–server application with a shared database and server-side identity and authorization. The SRS distinguishes these two states so that the prototype is not represented as already providing capabilities that are only planned.

2.2 Product context

The platform serves campus members who need a trusted, locally relevant place to reuse goods and students who need a structured path to internships, projects and mentorship. Recruiters need to publish opportunities and review candidates. Administrators need oversight, verification and moderation. The system is not intended to replace the university's official academic, HR or placement systems.

2.3 High-level functions

Area	Main capabilities
Shared platform	Navigation, account/profile, role-aware dashboards, notifications, theme, help and policy information
Marketplace	Browse/search/filter/sort listings, create/edit/delete own listings, favourites, seller enquiry, item status
Career Connect	Search opportunities, view details, apply with note, track status, recruiter role management, applicant review
Administration	User and organization verification, reports, content moderation, audit trail and configuration (target state)
Operations	Monitoring, backup, error logging, security controls, release and support

2.4 Design philosophy

The user experience shall be simple and task-oriented. Marketplace and Career Connect shall share a coherent visual system but retain clear module identity. The system shall not imply that local browser records are shared with other users. Any production migration must explicitly replace local-only persistence with a trusted backend and database.

2. Overall Description – Operating Environment

2.5 Operating environment

- Client devices: desktop computers, laptops, tablets and smartphones.
- Supported browsers: current and previous major versions of Chrome, Edge, Firefox and Safari, subject to testing.
- Minimum target viewport: 360 CSS pixels wide; layout shall reflow without horizontal page scrolling for primary workflows.
- Current prototype: browser with JavaScript enabled and localStorage available.
- Target deployment: HTTPS-enabled web hosting and server-side application environment; exact hosting provider remains an implementation decision.
- Network: campus Wi-Fi, mobile data and ordinary internet connections.

2.6 User environment

Most student users are expected to have basic web and smartphone literacy. Recruiters and administrators may use desktop browsers for reviewing multiple records. The UI shall use familiar terms, concise instructions, clear action labels and visible success/error feedback. A user shall not need specialist training to browse an item or apply for an opportunity.

2.7 Constraints

- The current academic prototype is browser-local and does not provide cross-device or cross-user synchronization.
- Seeded records are demonstration data and must be labelled or replaced before real deployment.
- No actual university identity verification, company verification, file upload, payment processing or real-time chat is present in the prototype as described by the conference presentation.
- The Lean Canvas identifies payment gateway and hosting as decisions to be finalized; payment is therefore not a committed baseline feature.

- Institutional policies and legal obligations must be confirmed before collecting real personal or recruitment data.

2.8 User documentation

The delivered system should include a short user guide, marketplace safety guidance, student application guide, recruiter guide and administrator/moderator guide. The help content shall explain eligibility, listing rules, application statuses, notification behaviour, privacy controls, reporting procedures and known limitations.

3. Stakeholders and User Classes

Stakeholder / class	Goals	Permissions / interaction
Guest visitor	Understand the platform and available opportunities/items	View public landing and permitted previews; no protected action
Student	Find affordable items; list reusable goods; save items; discover opportunities; apply and track progress	Manage own profile, listings, favourites and applications; message/enquire where enabled
Faculty / staff	Participate in campus reuse and share opportunities where permitted	Verified member functions subject to campus policy
Recruiter / company representative	Publish roles/projects, review applicants and update candidate progress	Manage own organization profile and postings; view applications to own postings
University administrator	Maintain trusted campus participation and policy compliance	Verify campus users, manage roles, review reports and suspend accounts
Moderator	Review suspicious listings/users and resolve policy reports	Access moderation queue and perform authorized actions
System operator	Maintain availability, backups and technical configuration	Infrastructure and operational privileges; no routine business moderation

3.1 Role principles

Roles shall be explicit and assigned through an authorized process. A student role shall not automatically confer recruiter or administrator permissions. A recruiter shall only access applications submitted to opportunities owned by that recruiter or organization. Administrative privileges shall be limited to designated personnel and recorded in audit logs.

3.2 Stakeholder consultation

Requirements shall be validated through walkthroughs with representative students, at least one faculty/staff representative, a recruiter persona and the project guide. Where real stakeholders are unavailable, the team shall document the persona assumptions and treat them as provisional rather than verified facts.

4. Product Scope and Release Boundaries

4.1 Release 0 – Demonstration prototype

- Single-page/single-file web prototype using HTML, CSS and JavaScript.
- Browser-side routing/state and localStorage persistence.
- Seeded marketplace listings and career opportunities.
- Marketplace search, category and condition filters, price sorting, create/edit/delete own listings, favourites and seller enquiry interaction as shown in the project deck.
- Career search by role/company/skills/type, recruiter role posting management, student applications with short note and status tracking.
- Shared dashboards, notifications and light/dark theme.

4.2 Release 1 – Campus pilot target

- Backend and centralized database for users, listings, opportunities, applications and notifications.
- Server-side authentication, university email verification and role-based authorization.
- Company/recruiter verification workflow.
- Secure image upload and moderation/reporting.
- Reliable in-app messaging or enquiry workflow with notification delivery.
- Audit events, backups, deployment, security testing and accessibility evaluation.

4.3 Release 2 – Candidate enhancements

- Marketplace rentals and exchanges with structured periods, reminders and offers, as defined in the Lean Canvas.
- Ratings and reviews after completed deals, report resolution and expanded moderation.
- Optional payment integration only after institutional, legal, security and operational approval.
- Multi-institution support, cross-campus search and shared reputation, subject to policy agreements.
- Mobile application/PWA enhancements, sustainability impact metrics and price guidance.

4.4 Scope control

A feature is not part of the current implementation merely because it appears in the Lean Canvas as a solution or future opportunity. In particular, rentals, exchange offers, in-app chat, ratings, admin console, rental reminders, push notifications and payments are target or proposed capabilities unless separately demonstrated in the current prototype. This separation is mandatory for accurate project reporting.

5. External Interface Requirements – User Interface

5.1 General UI requirements

- INT-UI-01: The system shall present a consistent header/navigation pattern across marketplace and Career Connect.
- INT-UI-02: The interface shall identify the active module and provide a clear route between modules.
- INT-UI-03: Forms shall label every input and communicate required fields and validation rules.
- INT-UI-04: Success, warning and error messages shall be visible and understandable without relying on colour alone.
- INT-UI-05: The interface shall support light and dark theme preferences; text and controls shall remain legible in both.
- INT-UI-06: Responsive layouts shall adapt cards, navigation, forms and tables to mobile, tablet and desktop widths.
- INT-UI-07: Destructive actions shall require confirmation or provide a reversible recovery mechanism where practical.

5.2 Main screens

Screen	Primary elements
Landing / Home	Product summary, module entry points, sign-in/account controls, featured or recent content
Marketplace	Search field, category and condition filters, price filter, sort control, listing cards
Listing detail	Title, description, category, condition, price, owner, status, images when enabled, save/enquiry actions
Create/edit listing	Title, category, condition, price, description, image selection in target state, validation and submit
Career opportunities	Search, role/company/skills filters, type selector, opportunity cards and details
Opportunity detail	Organization, role, type, eligibility, skills, description, deadline and apply action
Application dashboard	Applied opportunities, status, date and available next action
Recruiter dashboard	Owned opportunities, applicant totals, application review and status controls
Profile	Display name, education/affiliation, skills, contact visibility and account settings
Notifications	Event list, read/unread state and route to related item/application

5.3 Accessibility

The interface should target WCAG 2.1 Level AA. Controls shall be keyboard reachable, focus order shall be logical, images shall have meaningful alternative text, form errors shall be programmatically associated with fields and colour contrast shall be checked. Accessibility conformance must be tested; it shall not be claimed solely from visual inspection.

5. External Interface Requirements – Software, Hardware and Communications

5.4 Hardware interfaces

No dedicated hardware interface is required. The client device must support a modern browser, JavaScript and local storage for the current prototype. A camera or file picker may be used for listing/profile images in a later release, but such upload capability is not present in the described prototype.

5.5 Software interfaces

Interface	Purpose	Baseline status
Browser localStorage	Persist demo state in the current browser profile	Current prototype
Backend REST API	Centralized access to users, listings, opportunities, applications and notifications	Planned target
Database	Persistent shared system of record	Planned target; MongoDB is indicated in Lean Canvas technology stack
Email service	Verification, password reset and transactional notices	Provider TBD
Object storage	Store and serve listing/profile images	Planned target
Push notification service	Optional browser/device alerts	Proposed; provider TBD
University identity/email	Validate campus affiliation	Planned target; integration method TBD

5.6 Communications

- INT-COM-01: Target client-server traffic shall use HTTPS with TLS 1.2 or higher.
- INT-COM-02: Target API requests and responses shall use JSON unless a documented exception is approved.
- INT-COM-03: API errors shall use consistent status codes and structured error payloads.
- INT-COM-04: Real-time chat, if implemented, shall use WebSocket or an equivalent secure mechanism.
- INT-COM-05: Email notifications shall be sent through an approved transactional email provider.
- INT-COM-06: The client shall not store secrets, privileged credentials or server signing keys in browser code.

5.7 Browser storage boundary

The current localStorage implementation is suitable only for demonstration and single-browser persistence. It is not a security boundary, shared database or authoritative source of identity. Users can inspect or modify browser storage. Production permissions, verification, application status and listing ownership shall be enforced by the server.

6. Functional Requirements – Shared Platform: Accounts

6.1 Account registration

ID	Requirement	Priority	Verification
FR-SH-ACC-01	The system shall provide a registration form with the minimum fields approved for the selected user type.	Must	Test
FR-SH-ACC-02	The target system shall verify student/faculty/staff affiliation using an approved university email domain or institution-approved identity method before granting verified campus privileges.	Must	Test
FR-SH-ACC-03	The target system shall prevent duplicate accounts for the same normalized email address.	Must	Test
FR-SH-ACC-04	The system shall display registration validation errors adjacent to the relevant field.	Must	Inspection/Test
FR-SH-ACC-05	The system shall communicate whether registration is pending, verified, rejected or suspended.	Should	Test

6.2 Authentication and session

ID	Requirement	Priority	Verification
FR-SH-AUTH-01	The system shall allow registered users to sign in using the configured authentication method.	Must	Test
FR-SH-AUTH-02	The system shall provide a logout action that ends the active session.	Must	Test
FR-SH-AUTH-03	The target system shall provide a secure password reset or approved account recovery workflow.	Must	Test
FR-SH-AUTH-04	The target system shall enforce server-side authorization for every protected operation.	Must	Test
FR-SH-AUTH-05	The system shall show a clear message when an unauthenticated user attempts a protected action.	Must	Test
FR-SH-AUTH-06	The system shall not expose passwords or authentication tokens in page content, logs or user-visible errors.	Must	Inspection

Prototype note: the conference deck confirms a browser-based account experience but also states that accounts and records are stored locally. Local demo authentication must not be described as verified university authentication.

6. Functional Requirements – Shared Platform: Profiles and Dashboards

6.3 Profile management

ID	Requirement	Priority	Verification
FR-SH-PRO-01	The system shall allow a user to view their own profile.	Must	Test
FR-SH-PRO-02	The system shall allow a user to edit permitted profile fields such as display name, education/affiliation and skills.	Must	Test
FR-SH-PRO-03	The system shall validate profile fields and reject invalid lengths or formats.	Must	Test
FR-SH-PRO-04	The target system shall protect private email and contact information from public display unless the user explicitly authorizes disclosure.	Must	Test
FR-SH-PRO-05	The system shall display a user's own marketplace activity and career application summary in the dashboard.	Should	Test

6.4 Dashboard

The dashboard shall provide a consolidated view of the user's relevant activity without mixing private records from other users. The student dashboard may show personal listings, saved items, submitted applications and notification summaries. The recruiter dashboard may show the recruiter's own active roles, application counts and candidate status summaries.

ID	Requirement	Priority	Verification
FR-SH-DASH-01	The system shall display dashboard widgets based on the authenticated user's role.	Must	Test
FR-SH-DASH-02	The student dashboard shall link to the user's own listings, saved items and applications.	Must	Test
FR-SH-DASH-03	The recruiter dashboard shall link to the recruiter's own role postings and their applications.	Must	Test
FR-SH-DASH-04	Dashboard counts shall be derived from the current authorized records and shall not expose other organizations' private data.	Must	Test
FR-SH-DASH-05	The system shall provide an understandable empty state when the user has no activity.	Should	Inspection

6.5 Theme preference

The system shall provide light and dark theme selection. The selected theme should persist across reloads on the same device in the prototype and, in the target system, may be associated with the user profile. Theme selection shall not change business data or permissions.

6. Functional Requirements – Shared Platform: Notifications and Search

6.6 Notifications

ID	Requirement	Priority	Verification
FR-SH-NOT-01	The system shall present a notification area for relevant marketplace and career events supported by the release.	Must	Test
FR-SH-NOT-02	A marketplace notification shall identify the event type and provide a route to the related listing or enquiry where available.	Should	Test
FR-SH-NOT-03	A career notification shall identify the related application or opportunity and its latest status where available.	Must	Test
FR-SH-NOT-04	The system shall distinguish read and unread notifications.	Should	Test
FR-SH-NOT-05	The system shall avoid creating duplicate notifications for the same event and recipient.	Should	Test
FR-SH-NOT-06	Email and push delivery shall only be enabled after a provider and user preference policy are approved.	Could	Inspection

The Lean Canvas lists alerts for messages, rental due dates and offers. These are target marketplace events, not evidence that all are implemented in the current prototype. The current project deck specifically describes notifications for seller enquiries and application updates.

6.7 Search framework

ID	Requirement	Priority	Verification
FR-SH-SRCH-01	Search shall operate within the module selected by the user and shall not mix marketplace items with career opportunities unless a unified search mode is explicitly selected.	Must	Test
FR-SH-SRCH-02	The system shall trim leading/trailing whitespace and handle empty search input safely.	Must	Test
FR-SH-SRCH-03	The system shall show a clear no-results state and allow the user to revise filters.	Must	Test
FR-SH-SRCH-04	Search results shall be limited to records the user is authorized to view.	Must	Test

7. Functional Requirements – Campus Marketplace: Listings

7.1 Marketplace purpose

The marketplace supports campus reuse by allowing eligible campus members to discover and manage pre-owned goods. The Lean Canvas specifically identifies textbooks, electronics, hostel essentials, lab equipment, bicycles and sports items. The current conference deck shows Books, Electronics, Furniture, Cycles, Clothing and Stationery as sample categories. The category taxonomy shall be configurable so the team can preserve these examples without treating one list as final.

7.2 Listing creation and ownership

ID	Requirement	Priority	Verification
FR-MKT-LST-01	The system shall allow an authorized user to create a listing with a title, category, condition, price and description.	Must	Test
FR-MKT-LST-02	The system shall validate mandatory listing fields before saving or publishing.	Must	Test
FR-MKT-LST-03	The system shall associate each listing with its creator/owner.	Must	Test
FR-MKT-LST-04	A user shall be able to edit only listings they own, unless an authorized administrator acts under moderation policy.	Must	Test
FR-MKT-LST-05	A user shall be able to delete or deactivate only their own listing.	Must	Test
FR-MKT-LST-06	The system shall display listing status such as Active, Reserved, Sold or Removed when those states are enabled.	Should	Test

FR-MKT-LST-07	The system shall display a confirmation after a listing is successfully created, updated or removed.	Must	Test

7.3 Listing details

A listing detail view should display the title, description, category, condition, price, owner display identity, posted/updated date and current status. If image uploads are enabled in a later release, the detail view shall display approved images and accessible alternative text. The system shall not fabricate photos or condition claims.

7.4 Listing categories

The initial category set shall include at least Books/Textbooks, Electronics, Furniture/Hostel Essentials, Cycles, Clothing, Stationery, Lab Equipment, Sports Items and Other. Administrators may add or deactivate categories in a later release. A category change shall not orphan existing records; inactive categories shall remain readable on historical listings.

7. Functional Requirements – Marketplace: Discovery and Favourites

7.5 Browse and discovery

ID	Requirement	Priority	Verification
FR-MKT-SRCH-01	The system shall display available marketplace listings in a browse view.	Must	Test
FR-MKT-SRCH-02	The system shall support keyword search over listing title and, where available, description.	Must	Test
FR-MKT-SRCH-03	The system shall filter listings by category.	Must	Test
FR-MKT-SRCH-04	The system shall filter listings by condition.	Must	Test
FR-MKT-SRCH-05	The system shall allow a maximum-price filter; a minimum-price	Must	Test

	filter may be added in the target release.		
FR-MKT-SRCH-06	The system shall sort results by newest and price ascending/descending.	Must	Test
FR-MKT-SRCH-07	The system shall exclude inactive, sold or removed listings from default active results unless the user explicitly selects a historical/status filter.	Should	Test

7.6 Favourites / wishlist

ID	Requirement	Priority	Verification
FR-MKT-FAV-01	An authenticated user shall be able to save a listing to favourites.	Must	Test
FR-MKT-FAV-02	The user shall be able to remove a listing from favourites.	Must	Test
FR-MKT-FAV-03	The system shall provide a view of the user's saved listings.	Must	Test
FR-MKT-FAV-04	A favourite shall be associated with its user and listing in the target database.	Must	Test
FR-MKT-FAV-05	If a saved listing becomes unavailable, the system shall indicate its updated status.	Should	Test

7.7 Discovery usability

Filters and sort controls shall be clearly labelled and shall indicate active selections. Clearing filters shall restore the default result set. Search and filters shall be composable: applying a category filter shall not silently erase the keyword or price filter unless the user requests a reset.

7. Functional Requirements – Marketplace: Enquiries and Transactions

7.8 Seller enquiry

ID	Requirement	Priority	Verification
FR-MKT-ENQ-01	A user shall be able to initiate a seller enquiry from an available listing when authenticated.	Must	Test
FR-MKT-ENQ-02	The enquiry shall reference the listing and its owner.	Must	Test
FR-MKT-ENQ-03	The system shall notify the listing owner of a new enquiry in the supported notification channel.	Should	Test
FR-MKT-ENQ-04	The system shall prevent a user from sending an enquiry to themselves for their own listing.	Must	Test
FR-MKT-ENQ-05	The system shall preserve enquiry context so the owner can identify the listing being discussed.	Should	Test

7.9 Transaction lifecycle – target capability

The Lean Canvas describes buying, selling, renting and exchanging as the marketplace's core modes. The current conference deck demonstrates discovery, saving and seller enquiry; it does not claim complete transaction processing. The target design shall treat a transaction as an agreement between users, not as a verified payment unless a payment provider is formally integrated.

ID	Requirement	Priority	Verification
FR-MKT-TXN-01	The system shall allow users to record a transaction outcome for an item where the transaction workflow is enabled.	Should	Test
FR-MKT-TXN-02	The system shall support a seller marking a listing as Reserved or Sold.	Should	Test
FR-MKT-TXN-03	The system shall retain a transaction reference between the listing, participants and outcome.	Should	Test
FR-MKT-TXN-04	The system shall clearly state that physical handover and off-platform payment are user-arranged unless an approved payment feature is introduced.	Must	Inspection

7.10 Rentals and exchange – proposed extension

The Lean Canvas proposes fixed rental periods, due-date reminders, exchange offers and accept/reject/counter actions. These shall be treated as a planned extension and not as current prototype functions. If approved, the rental record must include owner, borrower, item, start date, due date, return state and reminder state. Exchange offers must identify both items, proposer, recipient, offer state and any counter-offer history.

7. Functional Requirements – Marketplace: Rentals, Exchange and Trust

7.11 Rental workflow – proposed

26. Owner enables the listing for rent and specifies the permitted rental period and any deposit/terms approved by policy.
27. Interested member sends a rental request with requested start and end dates.
28. Owner accepts, rejects or proposes an alternative period.
29. On acceptance, the system records the rental agreement and due date.
30. The system sends reminders before the due date and on overdue status according to approved notification rules.
31. Owner/borrower records return confirmation; the rental is closed or escalated for dispute handling.

ID	Requirement	Priority	Verification
FR-MKT-RENT-01	The target system shall support a rental request with start and due dates.	Could	Test
FR-MKT-RENT-02	The system shall prevent overlapping confirmed rental periods for the same item.	Could	Test
FR-MKT-RENT-03	The system shall send due-date reminders to both parties when reminders are enabled.	Could	Test
FR-MKT-RENT-04	The system shall record return confirmation and overdue status.	Could	Test

7.12 Exchange workflow – proposed

ID	Requirement	Priority	Verification
FR-MKT-EXC-01	A user shall be able to propose an exchange against an eligible listing.	Could	Test
FR-MKT-EXC-02	The recipient shall be able to accept, reject or counter the exchange proposal.	Could	Test
FR-MKT-EXC-03	The system shall retain the current state and history of exchange proposals.	Could	Test
FR-MKT-EXC-04	The system shall notify participants when an exchange offer changes state.	Could	Test

7.13 Ratings and reports – proposed

The Lean Canvas proposes ratings after a deal and reporting suspicious listings or users for admin review. These features require a verified transaction or interaction reference to reduce arbitrary ratings. Report categories, evidence retention, response timelines, appeal process and moderation sanctions must be defined by campus policy before release.

8. Functional Requirements – Career Connect: Opportunities

8.1 Purpose

Career Connect helps students discover structured early-career opportunities. The project deck identifies four opportunity types: Internship, Full-time, Capstone project and Mentorship. The system shall keep the opportunity type explicit so users can distinguish employment from academic or developmental opportunities.

8.2 Opportunity posting

ID	Requirement	Priority	Verification
FR-CAR-OPP-01	An authorized recruiter shall be able to create an opportunity posting.	Must	Test
FR-CAR-OPP-02	A posting shall include role/title, organization/company, opportunity type and description.	Must	Test
FR-CAR-OPP-03	A posting should include required skills, eligibility, location/mode and application deadline where relevant.	Should	Test
FR-CAR-OPP-04	The system shall associate each posting with its recruiter/organization owner.	Must	Test
FR-CAR-OPP-05	A recruiter shall be able to edit or remove only postings owned by that recruiter/organization.	Must	Test
FR-CAR-OPP-06	The system shall distinguish active, closed and draft opportunities in the target release.	Should	Test
FR-CAR-OPP-07	The system shall not allow applications to closed or expired opportunities.	Must	Test

8.3 Opportunity discovery

ID	Requirement	Priority	Verification
FR-CAR-SRCH-01	Students shall be able to search opportunities by role/title.	Must	Test
FR-CAR-SRCH-02	Students shall be able to search by company/organization.	Must	Test
FR-CAR-SRCH-03	Students shall be able to search by skills.	Must	Test
FR-CAR-SRCH-04	Students shall be able to filter by opportunity type.	Must	Test
FR-CAR-SRCH-05	The system shall provide an opportunity detail view before application.	Must	Test
FR-CAR-SRCH-06	The system shall show an understandable no-results state.	Must	Test

8. Functional Requirements – Career Connect: Applications

8.4 Student application

ID	Requirement	Priority	Verification
FR-CAR-APP-01	An authenticated student shall be able to apply to an active opportunity.	Must	Test
FR-CAR-APP-02	The application shall include a short applicant note or statement where required by the posting.	Must	Test
FR-CAR-APP-03	The system shall associate the application with the student, opportunity and organization.	Must	Test

FR-CAR-APP-04	The system shall prevent duplicate active applications by the same student to the same opportunity unless a reapplication policy allows it.	Must	Test
FR-CAR-APP-05	The system shall confirm successful submission and show the application in the student's dashboard.	Must	Test
FR-CAR-APP-06	The system shall prevent applications after the closing date or when the opportunity is closed.	Must	Test
FR-CAR-APP-07	The student shall be able to view the current status of each submitted application.	Must	Test

8.5 Application status model

Status	**Meaning**	**Allowed next states (target baseline)**
Submitted	Application received by the system	Shortlisted, Rejected, Withdrawn
Shortlisted	Recruiter has selected the application for further consideration	Interview, Rejected, Withdrawn
Interview	Interview or assessment stage is planned/in progress	Rejected, Offer/Selected, Withdrawn
Rejected	Application not proceeding	Terminal unless recruiter reopens under policy
Offer/Selected	Recruiter indicates positive selection outcome	Accepted, Declined, Closed
Withdrawn	Applicant has withdrawn the application	Terminal

The conference presentation explicitly lists Submitted, Shortlisted, Interview and Rejected as the current prototype status labels. Offer/Selected, Accepted, Declined and Withdrawn are recommended target-state extensions and require confirmation before implementation.

8.6 Status update

ID	Requirement	Priority	Verification
FR-CAR-APP-08	An authorized recruiter shall be able to update the status of an application to their own opportunity.	Must	Test
FR-CAR-APP-09	The system shall record the previous and new status, actor and timestamp for each status change in the target system.	Should	Test
FR-CAR-APP-10	The system shall notify the applicant when a recruiter changes the application status.	Must	Test
FR-CAR-APP-11	A student shall not be able to change a recruiter-controlled status.	Must	Test

8. Functional Requirements – Career Connect: Recruiter Management

8.7 Recruiter identity and organization

A recruiter account represents a person acting for an organization. In the target system, recruiter access shall be granted only after an approved verification process. The project artifacts do not specify the exact company verification method, so domain verification, document review or administrator approval remain open decisions.

ID	Requirement	Priority	Verification
FR-CAR-REC-01	The target system shall distinguish recruiter accounts from student accounts.	Must	Test
FR-CAR-REC-02	The system shall associate each recruiter with an organization profile.	Must	Test
FR-CAR-REC-03	A recruiter shall only view and manage applications for opportunities owned by that recruiter or organization.	Must	Test
FR-CAR-REC-04	The system shall provide a recruiter dashboard with counts for owned roles, applicants and shortlisted candidates.	Must	Test
FR-CAR-REC-05	The system shall allow a recruiter to review applicant profile information that the student has made available for recruitment.	Must	Test
FR-CAR-REC-06	The system shall not expose private student information beyond the approved application context.	Must	Test

8.8 Recruiter workflow

32. Recruiter signs in using an approved recruiter account.

33. Recruiter creates an opportunity and completes required fields.

34. The system validates and publishes the opportunity if the recruiter is authorized.

35. Students discover the opportunity and submit applications.

36. Recruiter opens the applicant list for the owned opportunity.
37. Recruiter reviews applications and updates statuses.
38. The system records the change and notifies the affected applicant.

8.9 Fairness and transparency

The system shall not automatically reject or rank applicants using hidden criteria. Any future ranking or AI-assisted matching feature must be separately specified, explainable, reviewed for bias and approved by the institution. The baseline product is an opportunity listing and application workflow, not an automated hiring decision engine.

9. Administration, Moderation and Audit

9.1 Administration scope

The Lean Canvas identifies an admin console for verifying users, reviewing reports and keeping audit logs. The current conference deck lists real identity/access, moderation/reporting and deployment readiness as next steps. Accordingly, administrative features are target-state requirements, not claims about the current local prototype.

ID	Requirement	Priority	Verification
FR-ADM-01	The system shall provide an administrator-only interface for authorized platform administration.	Must	Test
FR-ADM-02	The administrator shall be able to view user verification status and take approved verification actions.	Must	Test
FR-ADM-03	The administrator shall be able to assign or revoke roles according to an approved authorization policy.	Must	Test
FR-ADM-04	The moderator shall be able to view submitted reports in a moderation queue.	Should	Test

FR-ADM-05	The moderator shall be able to hide/remove a reported listing when policy grounds are met.	Should	Test
FR-ADM-06	The moderator shall be able to warn, restrict or suspend an account when authorized by policy.	Should	Test
FR-ADM-07	The system shall record administrative actions with actor, action, target, timestamp and reason.	Must	Test
FR-ADM-08	The system shall restrict audit-log access to authorized roles.	Must	Test

9.2 Reporting workflow

39. User selects Report on an eligible listing or profile.
40. System asks the user to select a reason and optionally provide a description/evidence.
41. System creates a report with status New and confirms receipt.
42. Moderator reviews the report and associated content.
43. Moderator records a decision: no action, warning, content restriction/removal or account action.
44. System records the decision and notifies the reporter where policy permits.
45. User may appeal if an appeal process is approved.

9.3 Audit requirements

Audit records shall be append-only for ordinary application users. They shall not contain passwords, raw authentication tokens or unnecessary sensitive information. Retention periods, access review and export rules shall be set by institutional policy.

10. Data Requirements – Conceptual Model

10.1 Core entities

Entity	Description	Key relationships
User	Account identity, role, status and profile	Owns listings, favourites, applications, notifications; may be recruiter
Organization	Company/recruiter identity and verification	Owns opportunity postings; associated with recruiter users
Listing	Marketplace item record	Owned by a user; may have images,enquiries,rental/ exchange /transaction records
Category	Marketplace classification	Referenced by listings
Favourite	User's saved listing relationship	Links user and listing
Enquiry / Conversation	Buyer-to-owner contact context	Links initiating user, listing and owner
Opportunity	Career role/project/mentorship posting	Owned by recruiter/organization; receives applications
Application	Student application to an opportunity	Links student, opportunity and status history
Notification	User-facing event alert	Belongs to recipient and optionally references an entity
Report	User-submitted moderation issue	Targets listing or user; reviewed by moderator
AuditEvent	Record of privileged or important state change	References actor and target

10.2 Data principles

- Each persistent record shall have a unique identifier.
- Creation and update timestamps shall be stored for important entities.
- Ownership and authorization relationships shall be explicit.
- Deleted or deactivated records shall be handled according to retention policy; hard deletion shall not destroy required audit evidence.
- The target database shall enforce uniqueness and referential integrity where supported.
- The current prototype's localStorage object model is not a final production data model.

10. Data Requirements – Data Dictionary: User and Organization

Field	Type / constraint	Description
user_id	UUID/ObjectId; unique	System-generated account identifier
display_name	String; 2–80 chars	Name displayed to other users
email	String; normalized; unique	Login/verification contact; private by default
role	Enum	Student, FacultyStaff, Recruiter, Moderator, Administrator
account_status	Enum	Pending, Active, Suspended, Deactivated
verification_status	Enum	Unverified, Pending, Verified, Rejected, Expired
institution	String / reference	Campus or institution affiliation
department_program	String; optional	Academic department or program
skills	Array of strings	Skills listed by a student or recruiter profile
created_at / updated_at	Timestamp	Record lifecycle timestamps
organization_id	Reference; optional	Organization association for recruiter
organization_name	String	Company or organization

		display name
organization_domain	String; optional	Domain used as one verification signal
organization_status	Enum	Pending, Verified, Rejected, Suspended

10.3 Data validation

Email addresses shall be normalized before uniqueness checks. Display names and descriptions shall be length-limited and escaped before rendering. Role and status fields shall be selected from controlled enumerations. The system shall not trust role values submitted by the browser; target authorization shall derive from the authenticated server-side identity.

10. Data Requirements – Data Dictionary: Marketplace

Field	Type / constraint	Description
listing_id	Unique identifier	Listing primary key
owner_user_id	User reference	Account that created the listing
title	String; 3–100 chars	Listing title
description	String; max length set by product policy	Item details and condition notes
category_id	Category reference	Controlled marketplace category
condition	Enum	New, Like New, Good, Used; final labels configurable
price_amount	Decimal >= 0	Asking price where sale is selected
price_currency	ISO currency code	Currency; INR for India pilot unless changed
listing_mode	Enum	Sell, Rent, Exchange, GiveAway; modes enabled by release

status	Enum	Draft, Active, Reserved, Sold, Inactive, Removed
image_refs	Array of object references	Approved image locations in target state
created_at / updated_at	Timestamp	Listing dates
favourite_id	Unique identifier	Favourite relationship
favourite_user_id	User reference	User saving listing
favourite_listing_id	Listing reference	Saved listing
enquiry_id	Unique identifier	Enquiry/conversation reference
enquiry_message	String	Initial enquiry content, if collected
enquiry_status	Enum	Open, Responded, Closed, Reported

10.4 Listing integrity

A listing shall have one owner and one active category. A listing cannot be both Sold and Active. If the owner is suspended or the listing is removed by moderation, the listing shall not appear in normal active search results. Price shall be non-negative and currency shall be explicit.

10. Data Requirements – Data Dictionary: Career and Notifications

Field	Type / constraint	Description
opportunity_id	Unique identifier	Opportunity primary key
recruiter_user_id	User reference	Posting owner
organization_id	Organization reference	Organization represented
title	String; 3–120 chars	Role/project/mentorship title
opportunity_type	Enum	Internship, Full-time, Capstone project, Mentorship
description	Text	Opportunity description
skills	Array of strings	Required/preferred skills
eligibility	Text / structured rules	Eligibility information
deadline	Date/time; optional	Application closing date

status	Enum	Draft, Active, Closed, Expired, Removed
application_id	Unique identifier	Application primary key
student_user_id	User reference	Applicant
application_note	Text; length-limited	Short note submitted by applicant
application_status	Enum	Submitted, Shortlisted, Interview, Rejected; target extensions require approval
status_history	Array of events	Previous/new status, actor and timestamp
notification_id	Unique identifier	Notification primary key
recipient_user_id	User reference	Notification recipient
event_type	Controlled enum	Enquiry, ApplicationSubmitted, StatusChanged, etc.
read_at	Timestamp; nullable	Read state

10.5 Data retention

Retention periods for inactive accounts, unsuccessful applications, reports, audit events and notifications must be approved by the institution before production. The SRS does not invent a statutory retention period. Users shall be informed about what profile and application data is visible to recruiters.

11. Business Rules and Workflow States

11.1 General business rules

Rule ID	Rule
BR-01	Only authorized account roles may perform role-restricted actions.
BR-02	A user may edit or delete only their own listing unless an administrator acts under policy.
BR-03	A recruiter may manage only opportunities and applications belonging to their organization or account.

BR-04	A student may apply only to an active opportunity before its deadline.
BR-05	The system shall not accept duplicate applications unless an approved reapplication policy applies.
BR-06	A marketplace listing marked Sold, Removed or Inactive shall not be offered as available.
BR-07	Private contact details shall not be publicly exposed by default.
BR-08	Payments and physical exchanges are user-arranged unless a separately approved feature is deployed.
BR-09	Demo/sample data shall be distinguishable from real user-generated data.
BR-10	Administrative actions shall be auditable.

11.2 Listing state model

Current state	Event	Next state
Draft	Owner publishes valid listing	Active
Active	Owner reserves item	Reserved
Reserved	Deal completes	Sold
Reserved	Reservation cancelled	Active
Active	Owner deactivates	Inactive
Any eligible state	Moderator removes for policy violation	Removed

11.3 Opportunity state model

Current state	Event	Next state
Draft	Recruiter publishes valid posting	Active
Active	Recruiter closes posting	Closed
Active	Deadline passes	Expired
Active	Moderator removes posting	Removed
Closed/Expired	Authorized recruiter reopens under policy	Active

11. Business Rules – Application and Interaction States

11.4 Application state model

The baseline states visible in the current project deck are Submitted, Shortlisted, Interview and Rejected. The target state model may include Withdrawn and Offer/Selected after stakeholder approval. State transitions shall be validated by the server and recorded in the application history.

From	To	Authorized actor	Rule
Submitted	Shortlisted	Recruiter for owning role	Application is reviewed and selected for next stage
Submitted	Rejected	Recruiter for owning role	Recruiter records rejection
Shortlisted	Interview	Recruiter for owning role	Interview stage is initiated
Shortlisted	Rejected	Recruiter for owning role	Candidate does not proceed
Interview	Rejected	Recruiter for owning role	Candidate does not proceed
Interview	Offer/Selected	Recruiter for owning role	Only if target extension approved
Any non-terminal	Withdrawn	Applicant	Only if withdrawal is enabled

11.5 Notification rules

- An application submission should create a notification for the recruiter/organization owner.
- A recruiter status change should create a notification for the applicant.
- A seller enquiry should create a notification for the listing owner.
- A notification shall link to an entity only if the recipient is authorized to view that entity.
- Notifications shall not disclose sensitive applicant information in lock-screen previews or email subject lines beyond approved policy.

11.6 Physical transaction safety

For future marketplace transactions, the platform should display safety guidance recommending public, well-lit campus meeting points and cautioning users not to share sensitive personal details. The platform shall not display precise private residential addresses. Dispute resolution and liability boundaries require institutional policy.

12. Non-Functional Requirements – Performance and Capacity

ID	Requirement	Measure / verification	Priority
NFR-PERF-01	For the campus pilot, common browse and dashboard pages should render their primary content within 3 seconds under the agreed test network and representative data volume.	Performance test	Should
NFR-PERF-02	Marketplace and career search should return results within 2 seconds for the agreed pilot dataset.	Performance test	Should
NFR-PERF-03	The target architecture shall support growth in users and records without redesigning the domain model.	Architecture analysis	Should
NFR-PERF-04	The system shall paginate or progressively load large result sets rather than render an unbounded list.	Test	Must
NFR-PERF-05	Image upload size and image count limits shall be configurable and validated before storage.	Test	Should
NFR-PERF-06	A failed notification provider shall not	Failure test	Must

	prevent a listing or application transaction from being saved.		

12.1 Performance test assumptions

The project artifacts do not provide measured user counts, peak concurrency or production traffic. The team shall define a pilot dataset and expected concurrent users before making numeric capacity claims. Performance figures in this SRS are proposed acceptance targets, not observed prototype results.

12.2 Scalability

The target system should separate presentation, business logic and persistence concerns. Database indexes should be considered for listing title/category/status, opportunity title/type/company, application student/opportunity/status and notification recipient/read state. Scaling decisions shall be based on measured demand.

12. Non-Functional Requirements – Reliability and Availability

ID	Requirement	Measure / verification	Priority
NFR-REL-01	The target system shall preserve successfully committed user records across normal application restarts.	Recovery test	Must
NFR-REL-02	The system shall provide clear failure feedback when an operation cannot be completed.	Test	Must
NFR-REL-03	A failure in a non-critical notification service shall not corrupt core marketplace or career records.	Fault injection	Must
NFR-REL-04	The target service should maintain a documented backup schedule and tested restoration procedure.	Inspection / restore test	Should
NFR-REL-05	The system shall avoid duplicate application or	Test	Should

	listing creation when a user retries after a network timeout, where an idempotency		

12.3 Availability

A monthly uptime objective may be established for a deployed pilot after hosting and operational support are selected. No availability percentage is claimed for the current local prototype. Planned maintenance, outages and support ownership shall be communicated to users.

12.4 Data recovery

The target service shall define recovery point and recovery time objectives with the institution. Backups shall be protected from unauthorized access and periodically tested. Restoring a backup shall not silently remove newer valid records without an incident recovery decision.

12. Non-Functional Requirements – Usability and Accessibility

ID	Requirement	Verification	Priority
NFR-USE-01	A first-time student shall be able to locate a marketplace listing and open its detail view without external help.	Usability test	Must
NFR-USE-02	A first-time student shall be able to locate an opportunity and submit a valid application using the provided instructions.	Usability test	Must
NFR-USE-03	Forms shall preserve valid entered values when a validation error occurs, unless security requires clearing them.	Test	Should
NFR-USE-04	The system shall provide empty states, loading states and error states for primary workflows.	Inspection/Test	Must
NFR-USE-05	The interface shall support keyboard navigation for primary actions.	Accessibility test	Must
NFR-USE-06	The interface shall use sufficient text/background contrast in light	Contrast	Must

	and dark themes.	analysis	
NFR-USE-07	Form controls shall have programmatically associated labels and accessible error descriptions.	Accessibility test	Must

12.5 Responsive behaviour

At narrow widths, navigation may collapse into a menu, cards may stack vertically and tabular recruiter data may become a card list or horizontally scrollable region with clear cues. No primary action shall become unreachable because of viewport width.

12.6 Language and content

The initial UI language is English. Text should avoid unnecessary technical jargon. The system may support additional languages in a later release; translations must preserve meaning and accessible labels.

12. Non-Functional Requirements – Maintainability and Portability

ID	Requirement	Verification	Priority
NFR-MNT-01	The target codebase should separate UI components, business rules, data access and configuration.	Code inspection	Should
NFR-MNT-02	The project shall use version control and documented setup instructions.	Inspection	Must
NFR-MNT-03	Critical business logic shall have automated tests; the team should target at least 70% line coverage for core logic when a test framework is established.	Coverage report	Should
NFR-MNT-04	Configuration values and secrets shall be kept outside committed source code.	Repository inspection	Must
NFR-PORT-01	The web application shall run on the supported browsers listed in Section 2.5.	Compatibility test	Must
NFR-PORT-02	The target system shall be deployable to a documented	Deployment test	Should

	hosting environment without changing business logic.		

12.7 Maintainability practices

- Use consistent naming and formatting conventions.
- Document setup, environment variables, build and deployment steps.
- Record API contracts and data model changes.
- Review code changes before merging into the main branch.
- Track defects and requirements changes in the project tracker.
- Keep sample data separate from production data.

12.8 Portability

The current localStorage prototype is browser-profile-specific. Moving to a centralized service changes persistence and authentication behaviour and shall be treated as a planned architectural migration, not a simple hosting operation.

13. Security, Privacy and Safety

13.1 Security objectives

The target system shall protect account identity, student applications, recruiter records, private contact information and moderation actions. Security controls shall be enforced server-side. Browser-side hiding of buttons or fields is not sufficient authorization.

ID	Requirement	Verification	Priority
NFR-SEC-01	All client-server communication shall use HTTPS with TLS 1.2 or higher.	Configuration/test	Must
NFR-SEC-02	Passwords, if locally managed, shall be stored only using a suitable salted one-way password hashing algorithm.	Inspection	Must
NFR-SEC-03	The target system shall enforce role-based access control on every	Penetration/authorization test	Must

	protected API operation.		
NFR-SEC-04	The system shall validate and sanitize untrusted input to mitigate injection and cross-site scripting.	Security test	Must
NFR-SEC-05	The system shall protect against cross-site request forgery where cookie-based sessions are used.	Security test	Must
NFR-SEC-06	Authentication and sensitive endpoints shall be rate-limited.	Security test	Should
NFR-SEC-07	Session expiry and logout shall invalidate the applicable session.	Test	Must
NFR-SEC-08	Secrets shall not be committed to source control or exposed in client bundles.	Inspection	Must

13.2 Privacy

- Collect only information required for account, marketplace and recruitment workflows.
- Explain what profile and application information recruiters can see.
- Keep personal email and phone details private by default.
- Provide a process for account deactivation and data correction, subject to retention obligations.
- Restrict administrative access and record privileged actions.
- Do not use real student data in demonstrations without authorization and informed consent.

13. Security, Privacy and Safety – Threats and Controls

Threat	Potential impact	Required control
Account impersonation	Untrusted user enters campus-only community	Approved email/identity verification; verification state; secure recovery
Broken access control	Student views another student's application or recruiter data	Server-side ownership checks and role-based authorization
Stored/reflected XSS	Malicious content executes in another user's browser	Output encoding, input validation, content security policy where feasible
Credential stuffing/brute force	Account compromise	Rate limiting, secure password storage, monitoring and recovery controls
Data exposure in localStorage	User can inspect or alter demo data	Clearly label prototype; do not store production secrets or sensitive records client-side
Fraudulent listing	Financial loss or unsafe meetup	Reporting, moderation, user guidance and verified accounts
Recruitment data leakage	Exposure of applicant information	Recruiter ownership scope, least privilege, audit logs and retention policy
Malicious file upload	Malware or stored script delivered through images	File type/size validation, content inspection, safe storage and controlled serving

13.3 Marketplace safety

The Lean Canvas positions trust and safety as a differentiator: university-email verification, ratings, reports, moderation and in-app chat are proposed controls. The current prototype does not yet provide all of them. A production release shall not advertise these protections until they are implemented and tested. Safe-meetup guidance should recommend public campus locations and warn users against sharing private addresses or credentials.

13.4 Security testing

Before a real campus pilot, the team shall test authentication, authorization boundaries, input handling, session management, data exposure, file handling (if enabled), dependency vulnerabilities and audit integrity. High-severity defects must be resolved before release.

14. System Architecture and Design Constraints

14.1 Current prototype architecture

The current deliverable described in the project presentation is a single HTML file with embedded styling and JavaScript. The interface handles navigation, forms, filters and state in the browser. localStorage stores accounts and records within that browser. This architecture is suitable for demonstrating interaction flows and visual behaviour, but it cannot provide shared campus records, authoritative identity verification or secure multi-user access control.

14.2 Target architecture

Layer	Responsibility	Candidate technologies / notes
Presentation	Responsive web UI, forms, cards, dashboards, theme	HTML, CSS, JavaScript and React are indicated in Lean Canvas; final framework selection to be confirmed
Application/API	Authentication, authorization, marketplace and career business rules	REST API; runtime/framework not finalized in Lean Canvas
Persistence	Users, organizations, listings, opportunities, applications, messages, reports	MongoDB (NoSQL) indicated in Lean Canvas
Real-time communication	Messaging and event updates	WebSocket indicated in Lean Canvas; not present in current prototype
Notification delivery	In-app, email and optional push	Push/FCM and SMTP are examples in Lean Canvas; provider TBD
Media storage	Listing/profile images	Cloud object storage; provider TBD
Security/operations	HTTPS/TLS, encrypted credentials, audit logs	Deployment and operational controls to be selected

14.3 Architectural constraints

- Do not couple core business rules to localStorage APIs.
- Keep identity, authorization and ownership validation on the server in the target system.
- Separate marketplace and career domain models while reusing shared account/profile/notification services.
- Use controlled configuration for institution email domains and organization verification policy.
- Design APIs so a future mobile client can reuse backend functions.

14. Architecture – Component Responsibilities and Deployment

14.4 Logical components

Component	Responsibilities
Account service	Registration, verification state, login, recovery, logout, role and profile
Marketplace service	Listing lifecycle, category, search/filter, favourites, enquiries, transaction state
Career service	Organization, opportunity lifecycle, application submission, status workflow
Notification service	Event generation, read/unread state and delivery integration
Moderation service	Reports, review queue, actions and appeals
Audit service	Append-only security and administrative event records
Media service	Upload validation, storage references and safe image delivery

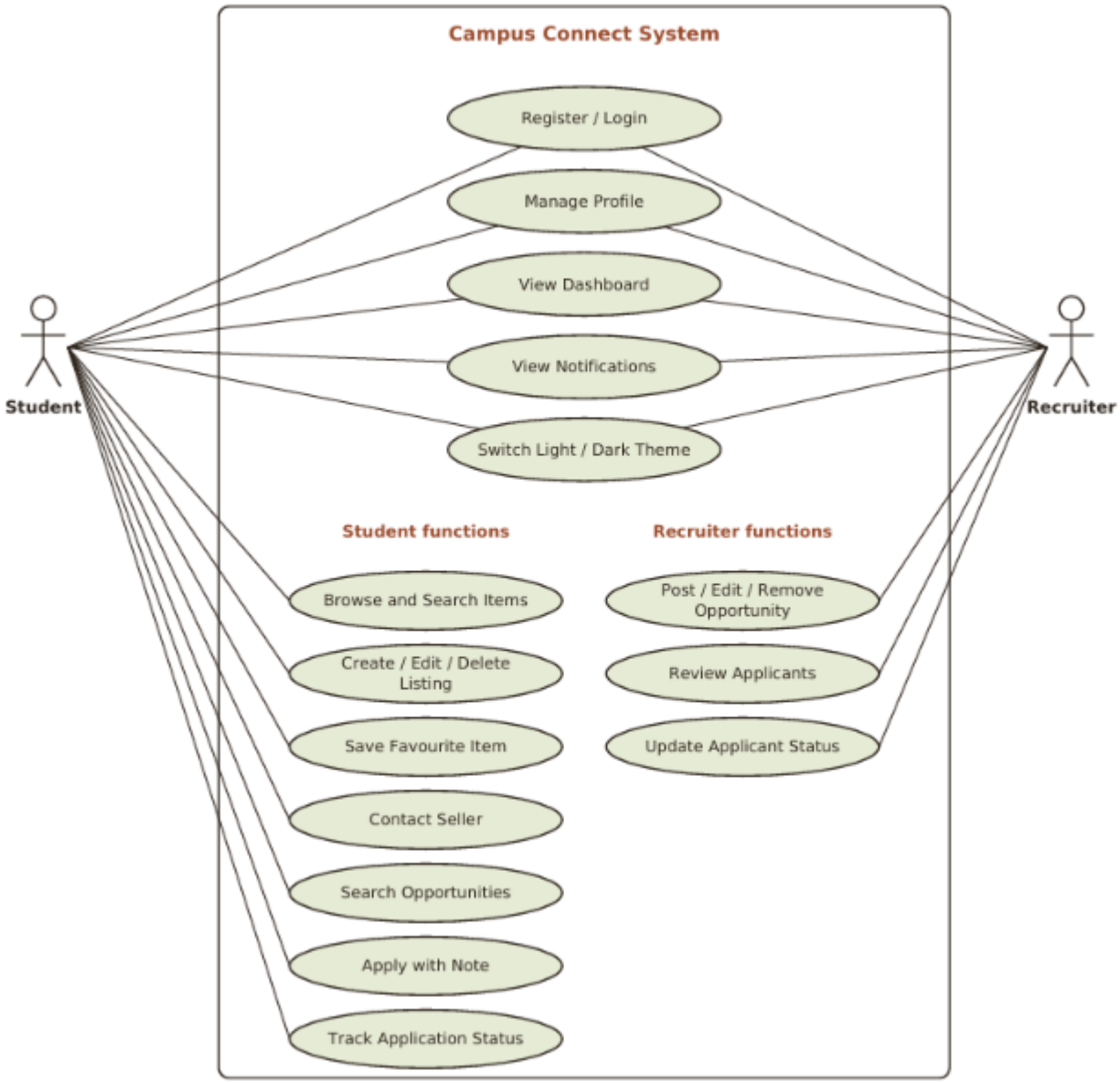
14.5 Deployment view

For the target release, the browser communicates with the application through HTTPS. The application communicates with the database and object storage using protected server-side credentials. Notification providers are external dependencies. Environment-specific configuration shall distinguish development, test and production. Production data shall not be copied into local development environments without authorization and sanitization.

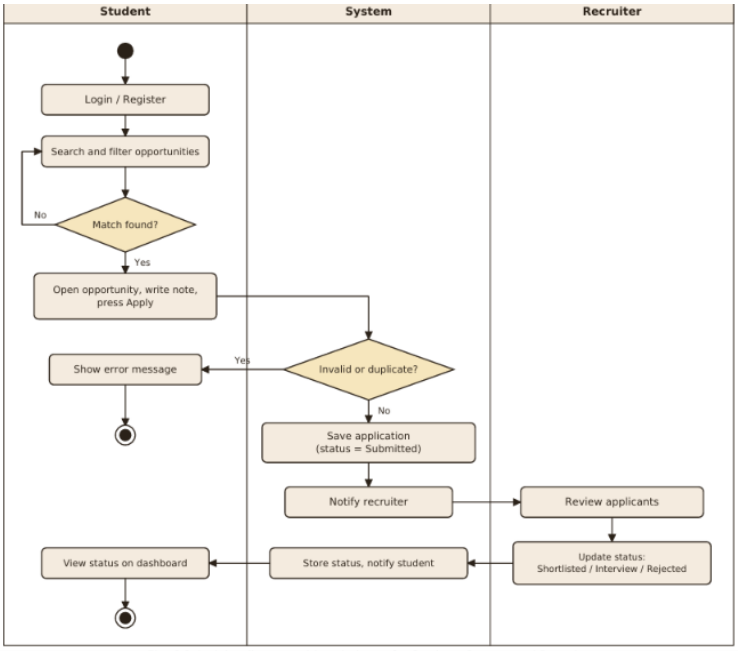
14.6 Integration boundaries

University email verification and company verification are external policy/integration boundaries. The SRS does not assume that a university directory API exists. The first implementation may use an approved email-domain verification workflow, with manual administrator review for exceptions. Integration selection shall be documented before implementation.

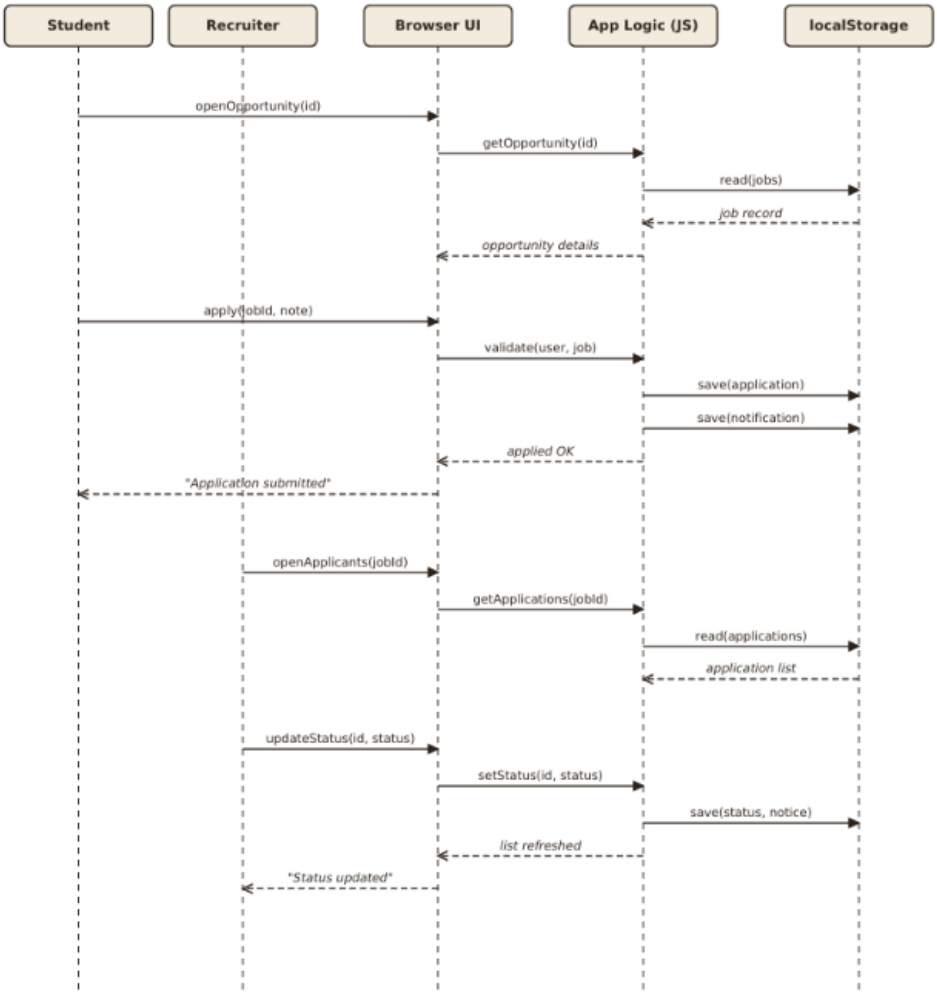
15. Use Cases – Marketplace



15.2- Activity Diagram



15.2- Sequence Diagram



16. Error Handling and Exceptional Conditions

16.1 General error behaviour

- Errors shall be written in plain language and explain the next useful action where possible.
- The system shall not expose stack traces, database details, secrets or internal server paths to end users.
- Validation errors shall identify the field and expected correction.
- Network and provider failures shall not be presented as successful operations.
- A failed save shall not show a success notification or create a phantom record.
- The system shall log sufficient diagnostic context without recording passwords or unnecessary personal data.

16.2 Marketplace exceptions

Condition	Expected response
Listing title missing	Prevent save and identify title field
Invalid/negative price	Prevent save; explain accepted numeric range
Listing owner mismatch	Reject operation; do not reveal private owner data
Listing no longer active	Prevent enquiry/transaction and display current status
Storage quota exceeded in prototype	Display storage warning and avoid silent data loss
Image upload rejected in target state	Explain file type/size restriction; do not publish unsafe file

16.3 Career exceptions

Condition	Expected response
Opportunity closed/expired	Disable application and show status
Duplicate application	Prevent duplicate or route to existing application
Recruiter not owner	Reject status update and log authorization failure
Invalid application note	Show validation and preserve valid fields
Notification service unavailable	Save application/status change; queue or retry notification

16.4 Recovery and resilience

Where an operation may be retried, the system should avoid duplicate creation. The UI shall distinguish a pending operation from a failed one. If the browser reloads during a local prototype action, the system should recover the last successfully persisted local state to the extent supported by localStorage.

17. Acceptance Criteria and Verification Strategy

17.1 Verification methods

Method	Meaning	Typical evidence
Test	Execute the system with controlled input and compare actual/expected results	Test case, result, screenshot/log
Inspection	Review code, configuration, UI or document	Checklist and reviewer notes
Analysis	Evaluate a model, metric, architecture or risk	Calculation/report
Demonstration	Show a user workflow to a reviewer	Recorded walkthrough or signed observation

17.2 Minimum acceptance scenarios

46. A student can open the marketplace and search by title/category/condition.
47. A student can filter by maximum price and sort by newest or price.
48. A student can create, edit and delete their own listing; a different user cannot edit it.
49. A student can save and remove a favourite and see saved items on the dashboard.
50. A student can search Career Connect by role, company, skills and opportunity type.
51. A student can submit a valid application and see Submitted status in the dashboard.
52. A recruiter can create and edit an owned opportunity and review applicants for that opportunity.
53. A recruiter can update an application status; the student can see the updated status.
54. The interface works at desktop and mobile viewport sizes and supports light/dark theme.
55. The prototype clearly communicates that local records are browser-local and not shared across users.

17.3 Pilot acceptance

A production/campus pilot shall additionally demonstrate server-side identity verification, role authorization, persistent shared records, privacy controls, moderation workflow, secure deployment, backup/restore and security test completion. These acceptance items cannot be passed by the current localStorage prototype.

18. Requirements Traceability Matrix – Shared

Requirement group	Source need / feature	Verification evidence
FR-SH-ACC / AUTH	Shared account registration and profile capabilities in project deck; target verification in Lean Canvas	Registration, verification, login, logout and role tests
FR-SH-PRO	Profile management and shared user experience in project deck	Profile edit/view and privacy tests
FR-SH-DASH	Marketplace dashboard and recruiter summary counts in project deck	Role-based dashboard tests
FR-SH-NOT	Seller enquiries and application updates in project deck; messages/rentals/offers in Lean Canvas	Event generation and recipient authorization tests
FR-SH-SRCH	Separate marketplace and career search in project deck	Search scope, no-results and access tests
FR-MKT-LST	Lean Canvas listing flow; project deck create/edit/delete	Listing CRUD and ownership tests
FR-MKT-SRCH	Project deck title/category/condition/price search and sorting	Filter composition and sorting tests
FR-MKT-FAV	Project deck save item/favourites	Add/remove and dashboard tests
FR-MKT-ENQ	Project deck seller enquiry; Lean Canvas chat direction	Enquiry association and notification tests
FR-MKT-RENT / EXC	Lean Canvas rent/exchange model	Future-release workflow tests
FR-CAR-OPP	Project deck recruiter posting management	Create/edit/remove opportunity tests
FR-CAR-APP	Project deck student apply with short note and status tracking	Application submit, duplicate and status tests

FR-CAR-REC	Project deck recruiter review and summary counts	Ownership and recruiter dashboard tests

18.1 Traceability rules

Every implemented feature shall map to one or more requirement IDs. Every requirement shall map to at least one verification case before release. If a feature is deferred, its requirement shall remain visible with a release label and shall not be marked complete.

18. Requirements Traceability Matrix – Quality and Administration

Requirement group	Source / rationale	Verification evidence
FR-ADM	Lean Canvas admin console, verification, reports and audit logs; project deck next-step moderation	Role access, report review and audit inspection
NFR-PERF	Responsive web product and expected campus browsing	Measured search/render performance under defined test dataset
NFR-REL	Target shared persistence and reliable workflow completion	Restart, backup/restore and notification-failure tests
NFR-USE	Student-friendly marketplace and career workflows	Representative user task completion and feedback
NFR-SEC	Verified campus trust proposition and recruitment privacy	Authorization, injection, session and data exposure tests
NFR-PORT	Browser-based responsive prototype	Cross-browser and viewport compatibility
DATA	Marketplace, career and shared dashboard records	Schema validation, ownership and consistency tests
INT	REST API, MongoDB, WebSocket, email/push, object storage direction in Lean Canvas	Integration and contract tests

18.2 Completion status vocabulary

Status	Meaning
Implemented – prototype	Demonstrated in the current browser-based prototype
Specified – target	Requirement documented but not yet demonstrated in the current prototype
Partially implemented	Some workflow elements exist; remaining behaviour is listed in the gap register
Deferred	Intentionally scheduled for a later release
Open decision	Cannot be finalized until policy, provider or stakeholder decision is made

19. Risks, Assumptions and Dependencies

ID	Risk / assumption	Impact	Mitigation / action
R-01	Campus identity verification method is not finalized	Trust boundary may be weak or delayed	Confirm approved email domains or identity integration with university
R-02	Company/recruiter verification method is not defined	Fraudulent or misleading opportunity posts	Define organization onboarding and approval process
R-03	Current prototype stores records locally	No shared user experience or authoritative records	Implement backend and centralized database before pilot
R-04	Moderation policy is TBD	Reports may be handled inconsistently	Define prohibited content, sanctions, appeals and response targets
R-05	Payment approach is undecided	User expectations and liability may be unclear	Keep payment off-platform in initial pilot; evaluate gateway separately
R-06	Rental return/dispute policy is undefined	Overdue items and disputes may not be resolved	Define terms, reminders, escalation and liability
R-07	Notification provider is TBD	Delivery may fail or incur cost	Select SMTP/push provider; implement retry and monitoring
R-08	No measured usage baseline	Performance targets may be unrealistic	Define pilot size and collect load-test data
R-09	Student application data is sensitive	Privacy and reputational risk	Apply least privilege, retention rules and access logging

R-10	Team capacity and schedule constraints	Feature scope may exceed delivery window	Prioritize core prototype and stage enhancements

19.1 Assumptions

- The first pilot is centered on Central University of Karnataka.
- Students and staff have access to an institutional email or an approved verification route.
- The system is initially English-language and browser-based.
- The current prototype is a demonstration baseline, not a live production service.
- The institution will review privacy, moderation and recruitment data handling before real deployment.

19.2 Dependencies

Backend runtime, hosting provider, database deployment, email provider, object storage, university verification route and recruiter verification policy are external decisions. The Lean Canvas mentions REST API, MongoDB, WebSocket, push/SMTP, cloud object storage and optional UPI/payment gateway; these are recorded as direction or examples rather than finalized procurement choices.

20. Future Scope and Release Plan

Phase	Scope	Exit criteria
Phase 0 – Prototype	Current two-module browser demo; localStorage; seeded listings/opportunities; core browse/list/apply flows	Demonstrate key user journeys and document limitations
Phase 1 – Campus pilot foundation	Backend/API, database, server authentication, university verification, recruiter verification, deployment	Shared persistent accounts and role-secure core workflows pass tests
Phase 2 – Trust and interaction	Image uploads, in-app messaging, notifications, reporting, moderation and audit logs	Safety/moderation workflows and secure media handling pass review
Phase 3 – Marketplace depth	Rentals, exchange offers, due-date reminders, ratings/reviews	Rental/exchange policy approved and workflows tested
Phase 4 – Expansion	Optional payment, multi-institution network, cross-campus listings, sustainability metrics, AI price suggestions	Governance, compliance, support and scalability approved

20.1 Lean Canvas-aligned growth

The Lean Canvas proposes a CUK pilot first, followed by rental/exchange refinements and optional payments, and later a multi-university network linking universities, NITs and IITs. This SRS preserves that staged direction. Multi-institution support is not part of the first pilot baseline.

20.2 Success metrics

- Verified active users.
- Number of active listings and completed deals.
- Rental on-time return rate once rentals are implemented.
- Exchange offers accepted once exchange is implemented.
- Student opportunity applications and status progression.
- Recruiter role postings and applicant review activity.
- User satisfaction, task completion and moderation resolution time.

Metric definitions, reporting windows and data collection consent shall be documented before these measures are used to claim impact.

Appendix A – Lean Canvas Reference Analysis (Marketplace)

A.1 Role of the Lean Canvas in this SRS

The submitted Lean Canvas is a one-page business/product model for the Campus Connect second-hand marketplace. It is used here to inform the marketplace requirements, target users, value proposition, channels, metrics and staged roadmap. It does not replace the wider product scope documented in the current Campus Connect technical presentation, which explicitly describes two connected experiences: Campus Marketplace and Career Connect.

A.2 Problem and customer segments

The Lean Canvas identifies four marketplace problems: difficulty finding affordable textbooks, lab and hostel items; lack of an organized resale/exchange path after a term; unsearchable and unverified notice boards/chat groups; and lack of a trusted fixed-period rental process. It identifies students as the primary users, faculty/staff as additional users and administrators as moderation users. Early adopters include first-year and final-year students around term start/end.

A.3 Solution and value proposition

The proposed solution is a verified campus second-hand market with buy/sell/rent/exchange, chat, ratings, reports, admin console, rental reminders and exchange offers. The value proposition is cheaper study material, less waste through reuse, safer deals through ratings/moderation, and renting or swapping instead of buying new. The SRS converts these into explicit requirements while clearly labelling rentals, ratings, chat and moderation as target capabilities where not in the current prototype.

A.4 Unfair advantage and channels

The Canvas identifies a closed email-verified community and the team's lived experience as students as the unfair advantage. Acquisition channels include university email/orientation, hostel and department groups, QR posters, student clubs and campus notice boards. These are launch/adoption strategies rather than software functions; the SRS captures their implications for verification, onboarding and usability.

Appendix A – Lean Canvas Analysis (Metrics, Costs and Risks)

A.5 Key metrics translated into product measures

Lean Canvas metric	Operational definition proposed for measurement	Related requirement area
Verified active users	Distinct verified accounts active within a defined reporting period	Account verification / analytics
Listings and completed deals	Count of active listings and deals with recorded completion state	Marketplace listing / transaction
Rental on-time return rate	Rentals returned on or before due date divided by rentals due in period	Rental workflow – future
Exchange offers accepted	Accepted exchange offers divided by submitted offers in period	Exchange workflow – future

A.6 Cost structure and revenue assumptions

The Canvas lists cloud hosting and image storage, push notifications and SMTP, payment gateway fees (optional), and development/moderation as cost categories. It proposes free use for verified students, featured listings, a small in-app payment fee and campus-partner sponsorships as possible revenue streams. These are business assumptions and not commitments to charge students. Any fee, sponsorship, payment or monetization feature requires explicit institutional and user-policy approval.

A.7 Alternatives and differentiation

The Canvas compares Campus Connect with WhatsApp/Telegram groups, notice boards and generic marketplaces. Its differentiation claims are campus-only verification, second-hand buy/sell, rentals/exchange with reminders, searchable filters/ratings and a planned multi-institution network. The requirements emphasize verifiable capabilities rather than comparative marketing claims.

A.8 Challenges

- Initial trust and adoption.
- Listing moderation and disputes; policy is not yet defined.
- Data privacy and compliance.
- Payment gateway and hosting selection.
- Campus account verification.
- Rental return and due-date enforcement.

These challenges are reflected in the risks, open decisions, security requirements and staged release plan. No assumption is made that the Lean Canvas resolves these governance questions.

Appendix B – Prototype-to-Target Gap Register

Capability	Current prototype status per project deck	Target requirement / gap
Marketplace browsing/search	Demonstrated: title, category, condition; max price; sort newest/price	Add backend search, pagination, robust filters and access rules
Marketplace listing management	Demonstrated: create, edit and delete personal listings	Add server ownership checks, image uploads, moderation and lifecycle
Favourites	Demonstrated: save item	Centralize per-user favourites and handle unavailable listings
Seller contact	Seller enquiry interaction shown	Implement persistent in-app messaging, privacy controls and notifications
Rent and exchange	Lean Canvas proposal; not listed as current prototype capability	Define rental terms, dates, exchange offer state and dispute process
Ratings and reports	Lean Canvas proposal; not current prototype feature in deck	Define verified transaction basis, moderation queue and appeals
Career opportunity search	Demonstrated: role, company, skills and type	Add server-backed filtering, deadlines and eligibility
Student applications	Demonstrated: short note and status tracking	Add secure persistence, duplicate prevention, status history and notifications
Recruiter management	Demonstrated: create/edit/remove roles, review applicants, update status	Add recruiter verification, organization ownership and audit
Accounts	Browser-stored prototype accounts/records	Replace with secure server-side authentication and verification
Notifications	Prototype notifications for seller enquiries/application updates described	Add durable event processing, delivery preference and retries
Admin console	Lean Canvas target; not current prototype feature in deck	Build user verification, reports, moderation and audit
Payments	Optional/proposed; not in prototype	Keep off-platform until separately approved
Multi-institution	Future scope	Tenant model, federation and cross-campus policy

This gap register is the principal bridge between the present demonstration and the complete target requirements. It prevents future-state features from being misreported as already implemented.

Appendix C – Sample Test Scenarios

Test ID	Scenario	Expected result
TC-MKT-01	Search a known listing title	Matching listing appears; unrelated records do not
TC-MKT-02	Apply category and condition filters	Only records matching both filters appear
TC-MKT-03	Set maximum price	Listings above maximum are excluded
TC-MKT-04	Sort by price ascending	Visible results are ordered lowest to highest
TC-MKT-05	Edit another user's listing	Operation is denied in target system
TC-MKT-06	Save then remove favourite	Favourite appears then disappears from saved list
TC-CAR-01	Search by opportunity type	Only selected type appears
TC-CAR-02	Apply to active opportunity	Application is saved with Submitted status
TC-CAR-03	Apply twice to same opportunity	Duplicate is prevented or existing application is shown
TC-CAR-04	Apply after deadline	Application is rejected with clear explanation
TC-CAR-05	Recruiter changes own applicant status	Status updates and applicant is notified
TC-CAR-06	Recruiter attempts to view another company's applications	Access is denied
TC-SH-01	Switch light/dark theme and reload	Theme preference persists as supported
TC-SH-02	Open dashboard with no activity	Empty state appears without error
TC-SEC-01	Submit script-like text in listing description	Text is rendered safely and does not execute
TC-SEC-02	Access protected endpoint without session	Request is rejected
TC-ADM-01	Moderator removes reported listing	Listing is hidden and action is audited

C.1 Test data

Use fictional users, fictional organizations, sample listings and test opportunities. Do not use real student application details or personal contact information. Test accounts shall represent student, recruiter, moderator and administrator roles.

Appendix D – Glossary, Acronyms and Open Decisions

Term	Definition
API	Application Programming Interface
Campus member	Student, faculty or staff member eligible to use campus-restricted functions
Career Connect	Campus Connect module for opportunities, applications and recruiter workflows
Enquiry	A buyer's initial contact with the owner of a marketplace listing
Favourite / Wishlist	A saved listing associated with a user
Listing	A marketplace record describing an item offered for sale, rent, exchange or giveaway
localStorage	Browser storage mechanism used by the current prototype; not a shared database
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
Opportunity	Internship, full-time role, capstone project or mentorship posting
Recruiter	Authorized company/organization representative managing opportunities and applications
SRS	Software Requirements Specification
TLS	Transport Layer Security
UVP	Unique Value Proposition

D.1 Open decisions requiring confirmation

Decision	Why it matters	Owner / next action
Approved university email domains / identity process	Defines verified campus membership	Confirm with university / guide
Recruiter/company verification	Prevents fraudulent job postings	Define administrator approval policy
Backend runtime and API framework	Lean Canvas says runtime to be finalized	Team architecture decision
MongoDB deployment and hosting	Persistence, cost and operations	Team / guide approval
Email and push provider	Verification and notification delivery	Compare providers and approve
Image upload limits and storage	Performance, safety and cost	Set file type, size, count and retention
Marketplace transaction/rental terms	Disputes, deposits and return enforcement	Define campus policy before rentals
Moderation policy and appeal	Fair and consistent enforcement	Draft and approve rules
Application status extensions	Offer/withdrawal stages not in current deck	Validate with recruiter/student stakeholders
Data retention and privacy notice	Personal and recruitment data obligations	Institutional review
Payment integration	Financial/security/legal responsibilities	Keep out of first pilot pending approval
Pilot scale and performance thresholds	Needed for credible capacity targets	Agree test dataset and concurrency

D.2 Final scope statement

Campus Connect is specified as a two-module web platform comprising Campus Marketplace and Career Connect. The Lean Canvas informs the marketplace module and its business assumptions; it does not narrow the overall product to marketplace-only. The current implementation is a local browser prototype. The target campus-ready system requires centralized persistence, verified identities, secure authorization, richer interactions, moderation and deployment controls. Features not present in the prototype are explicitly identified as target, proposed or deferred.